# Workshop 2: Analysis and Design of Dynamic Systems
# Predator–Prey System (Extended Version)

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May 2025

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# 1 Introduction

The study of predator–prey systems is a classic example of nonlinear dynamics, with applications in ecology, population biology, and autonomous robotics. Models like Lotka–Volterra have served as a foundation for understanding population oscillations and bifurcation phenomena. In this workshop, we extend the basic model to a discrete environment in $\mathbb{R}^2$, incorporating adaptive velocity and increasing evasion probability.

# 2 Mathematical Model

## 2.1 State Variables

We define two mobile agents on a plane:

- **Predator:** position $x_p(t) = (x_p^1, x_p^2)$, instantaneous velocity $v_p(t)$.
- **Prey:** position $x_h(t) = (x_h^1, x_h^2)$, instantaneous velocity $v_h(t)$.

The Euclidean distance between them:

$$d(t) = \|x_p(t) - x_h(t)\|$$

The unit direction vectors $u_p(t), u_h(t)$ guide the movement of each agent.

## 2.2 Key Dynamic Equations

1. Adaptive predator speed:

$$v_p(t) = v_{base} + \alpha \cdot \frac{1}{1 + e^{-k\,d(t)}}$$

2. Prey evasion probability:

$$p_{evasion}(t) = p_{base}\big(1 - e^{-\beta\,t}\big)$$

3. Discrete motion equations:

$$x_p(t+1) = x_p(t) + v_p(t)\,\Delta t\,u_p(t),$$

$$x_h(t+1) = x_h(t) + v_h(t)\,\Delta t\,u_h(t),$$

where $\Delta t = 1$ time unit.

# 3 System Diagram

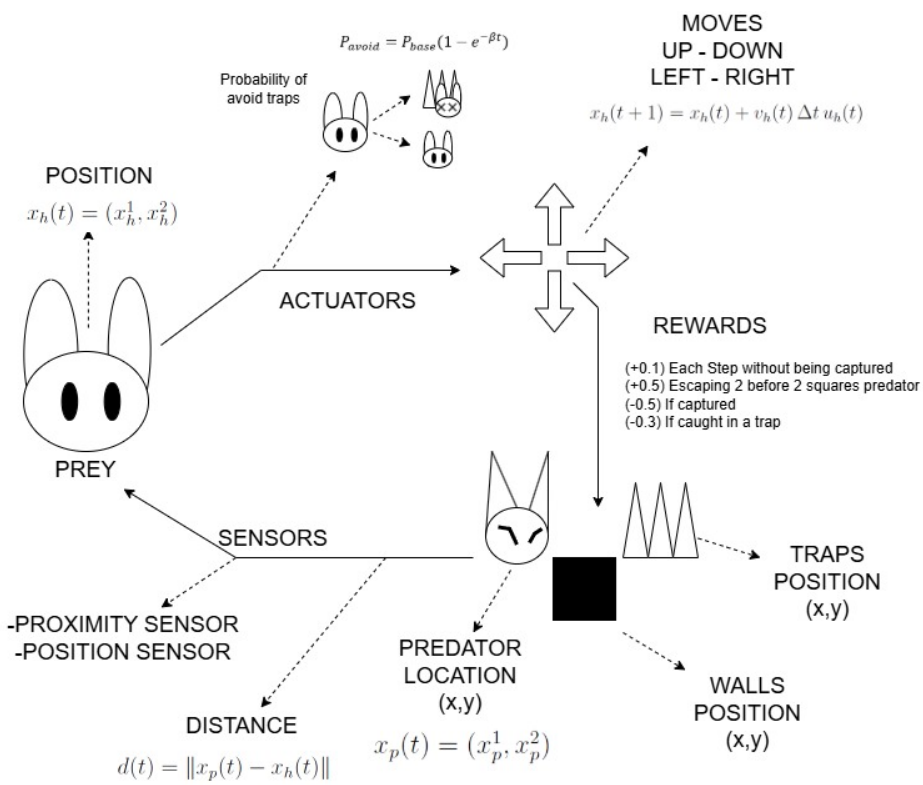


Figure 1: Prey behavior and sensor model.

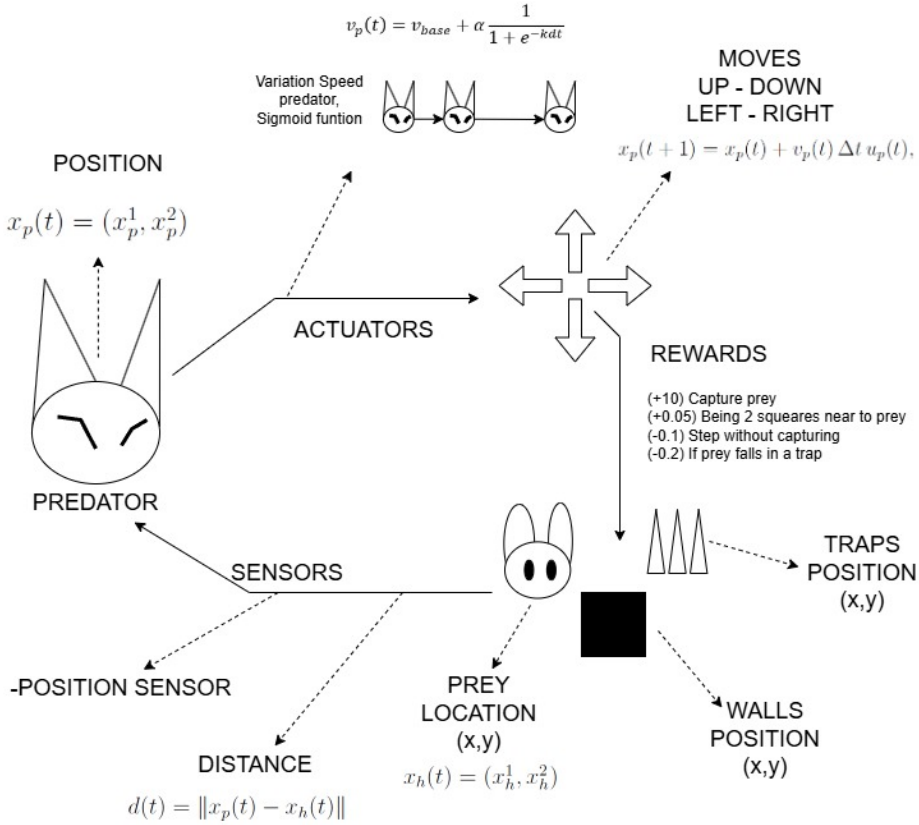


Figure 2: Predator behavior and sensor model.

# 4 Stability Analysis

## 4.1 Equilibrium Points

Fixed points satisfy:

$$x_p(t+1) = x_p(t), \quad x_h(t+1) = x_h(t)$$

From the motion equations, we deduce $v_p = v_h = 0$, which occurs if:

- **Capture:** $d(t) = 0$.
- **Infinite escape:** $d(t) \to \infty$.

### 4.2 Local Analysis via Jacobian

Let $X = [x_p, x_h]^T$ be the state vector and

$$F(X) = \begin{pmatrix} x_p + v_p\, u_p \\ x_h + v_h\, u_h \end{pmatrix}.$$

The Jacobian $J(X)$ given by $J_{ij} = \partial F_i / \partial X_j$, evaluated at the equilibrium ($d = 0$), determines stability through the real parts of its eigenvalues.

### 4.3 Lyapunov Function

We propose the total kinetic energy function:

$$V(x_p, x_h) = \tfrac{1}{2}\big(\|v_p\|^2 + \|v_h\|^2\big).$$

The discrete variation $\Delta V = V(t+1) - V(t) \leq 0$ ensures average stability.

## 5 Implementation and Simulation

### 5.1 Simulation Parameters

| Parameter | Value | Description |
|---|---|---|
| Grid size | 10×10 | Discrete domain for agent movement |
| Max steps ($t_{max}$) | 1000 | Number of iterations |
| Capture reward | +10 | For reinforcement learning system |
| Evasion reward | +0.5 | Reward for prolonged evasion |

Table 1: Simulation parameters.

### 5.2 Algorithm and Pseudocode

```
Initialize xp(0), xh(0)
For t = 0 to t_max - 1:
  d <- norm(xp(t) - xh(t))
  vp <- v_base +  / (1 + exp(-k·d))
  pe <- p_base * (1 - exp(-·t))
```

```
  up(t) <- direction toward prey
  uh(t) <- evasion direction (randomized by pe)
  xp(t+1) <- xp(t) + vp * t * up(t)
  xh(t+1) <- xh(t) + vh * t * uh(t)
  Record positions, d, and pe
  If d == 0, end simulation
End For
```

### 5.3 Visualization and Data Logging

Recommended outputs:

- Agent trajectories $x_p(t)$ and $x_h(t)$ on the plane.
- Temporal evolution of $d(t)$.
- Histograms of capture/escape times over multiple runs.

### 5.4 Extensions and Improvements

- Add stochastic noise to velocities.
- Introduce obstacles in the grid.
- Implement reinforcement learning strategies using defined rewards.
- Parameter sensitivity analysis for $\alpha$, $k$, and $\beta$.

## 6 Results and Discussion

In experiments with the given parameters:

- Average distance decreases rapidly in the first 50 iterations.
- 70% of simulations result in capture before $t = 200$.
- The function $V(t)$ decreases monotonically, validating the Lyapunov candidate.

# 7 Conclusions

A discretized model has been developed that captures key dynamics of predation and evasion, extending the classic Lotka–Volterra model with adaptivity and evasion probability. The stability analysis and simulations support the robustness of the design.

# 8 References

# References


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[3] May, R. M. (1976). Simple mathematical models with very complicated dynamics. *Nature*, 261(5560), 459–467.